

Computing CMB Temperature Fluctuations for Spherical Spaces

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Manifolds & Homotopy Groups

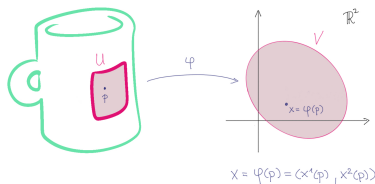


Figure: The prototypical example of a manifold – a mug [13].

Manifolds & Homotopy Groups

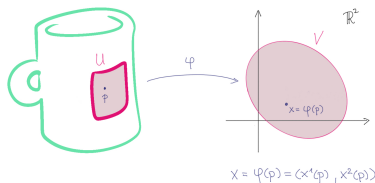


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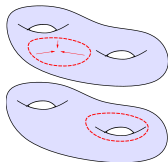


Figure: Two double tori with (non)-contractible paths [7].

Quotients of the 3-sphere

- The group $SO(4)$ is isomorphic to the isometry group of $\mathbb{S}^3$.
- Subgroups $\Gamma \leq SO(4)$ define **equivalence classes** of orbits on $\mathbb{S}^3$ by the standard action of $SO(4)$ on $\mathbb{R}^4$, i.e. $x \sim y$ iff $x = My$ for some $M \in \Gamma$.

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- The obtain space is (*sometimes*) a manifold. In particular, the finite Γ we will consider guarantee the manifold to be well defined and spherical.
- This can be easily generalized to the n -sphere.

Lens Spaces

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Definition (Lens space)

Given coprime $p, z \in \mathbb{Z}$, we define

$$L(p, q) := \mathbb{S}^3 / \langle M \rangle,$$

where $\langle M \rangle$ is the group generated by

$$M := \begin{bmatrix} \cos(2\pi/p) & \sin(-2\pi/p) & 0 & 0 \\ \sin(2\pi/p) & \cos(2\pi/p) & 0 & 0 \\ 0 & 0 & \cos(2\pi q/p) & \sin(-2\pi q/p) \\ 0 & 0 & \sin(2\pi q/p) & \cos(2\pi q/p) \end{bmatrix}.$$

The Laplace-Beltrami operator

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a generalization of the Laplace operator to more general spaces

$$\Delta f := \operatorname{div} \nabla f.$$

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- 1 The spectrum of this operator is a subset of $[0, \infty)$. We call it the *spectrum* of the manifold.
- 2 The spectrum determines many things about the space (like its volume).
- 3 When two distinct (up to isomorphism) manifolds share a spectrum, they form an *isospectral pair*.

The silver lining

- Isospectral manifolds are intriguing in their own right, but in dimension 3 all isospectral manifolds are isometric.
- We can thus attempt to infer the shape of our universe based on its spectrum.

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- The set of all possible wave numbers [for which a nonzero solution exists] depends on Γ - usually a proper subset of $\mathbb{N}$.
- Solutions to the Helmholtz equation are calculated for each Γ and CMB anisotropy $\frac{\delta T}{T}$ computed as a sum of spherical harmonics.

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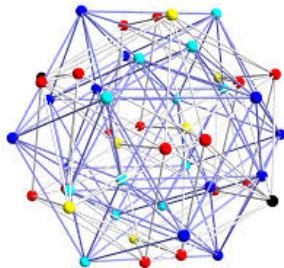


Figure: Graphical representation of the binary tetrahedral group [5]

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- Manifolds of the form $\mathbb{S}^3/\Gamma$ with Γ a group acting on $\mathbb{S}^3$.
- **Multi-connected** space: it has non-contractible loops.
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 - ① homogeneous: $N3$ and $L(8, 1)$.
 - ② inhomogeneous: $N2 \equiv L(8, 3)$.
- Results: inhomogeneous spaces have **more variety** in the CMB anisotropies than homogeneous spaces, because the strength of anisotropy suppression is **observer dependent**.

Statistical Isotropy and Hypothesis

Theoretical Expectation:

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Key Question: Does the observed CMB data deviate from statistical isotropy? If $\langle a_{\ell m} \rangle \neq 0$, this suggests a preferred cosmic direction.

Methodology: Sky Masking and the Test Statistic

Challenges in Real Observations:

- We cannot observe the full CMB sky due to foreground contamination.
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Monte Carlo Simulations:

- We generate thousands of **simulated CMB skies** assuming isotropy.
- Each sky has different random $a_{\ell m}$, drawn from:

$$a_{\ell m} \sim \mathcal{N}(0, C_\ell).$$

- The observed WMAP data is compared against these simulations.

Results and Interpretation

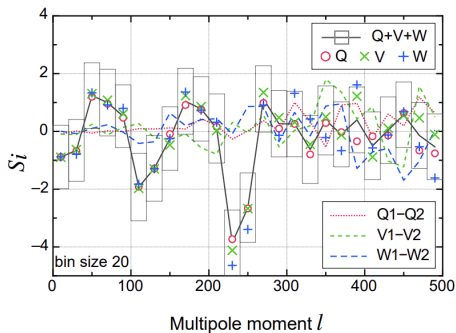
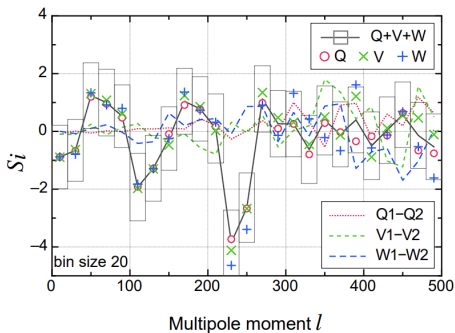


Figure: The decorrelated band mean test statistic values over multipole ranges [9].

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Key Findings:

- A statistically significant deviation was found in $221 \leq l \leq 240$.
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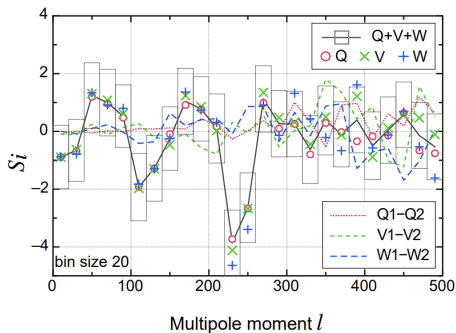


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Possible Explanations:

- A real cosmological signal? → A **finite universe** or new physics.
- A systematic effect? → Technological or theory-related limitations.

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- ② There are two homogeneous spherical manifolds obtained as $\mathbb{S}^3/\Gamma$ which produce CMB similar to observations.
- ③ Inhomogeneous spherical spaces exhibit varied behavior of CMB anisotropies.
- ④ Statistical test results suggest possibilities of **finite multi-connected** topology.

References

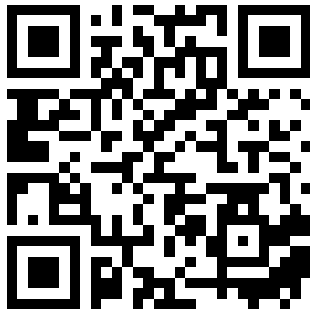
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💫 Thank You! 💫